Contextual Intelligence: A Causal Layer for Country-Specific Decisions

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Abstract

This paper describes a system that began with a conversation in a Himachal village in September 2024, by the founder's account, and became a country-specific causal layer for economic events. Unexpected rain had cut the apple harvest. The founder traced what a short harvest does next, found that India imports apples from Australia and New Zealand, and saw that a person who understood the chain had time to act before the market did. The same reasoning, repeated across events, produced the central observation of this work: an event is not good or bad. It moves gain and loss between sectors and people, with delays, and differently in different places.

We argue that general language models are necessary and not sufficient for decisions of this kind. They supply language and reasoning. They do not supply a country's sector structure, policy channels, calendar, or current state, and they cannot say when they do not know. We describe a three-part stack: the general model; a contextual layer made of an event taxonomy, a rule layer, primitive economic pressures, sector propagation, a refusal branch, a human review loop, and an outcome monitor; and the human who approves every change to that layer's knowledge. Each part is necessary and none is sufficient.

The layer's product is one structured object per event: the event, a signed vector across 32 sectors, stakeholder views, market regime, and audit data. The object is written once and read many ways, by a developer's program, a dashboard, or a purchase manager. India is the first implementation. The architecture is the same for any country; the content is not, and each country must write its own. We document the architecture as built, the dated record of its move from a static sentiment model to daily ingestion, the evaluation framework, and the limits. The claim concerns incidence, not price prediction.

We present this as a design pattern, not a product. The pattern is what transfers; the content is what each country and each industry must write for itself. This paper presents a systems framework and a development record. It does not present a benchmark result. The framework's first research priority is a signed sector-redistribution benchmark, which does not yet exist.

Keywords: causal incidence, country-specific AI, contextual layer, financial event classification, refuse-and-review

JEL codes: G14, G17, C55, D83, C45

Declaration of interest: The author is the founder of AION Analytics, whose system is described in this paper. Statements marked as the founder's account are the author's own, recorded here for provenance.

1. Introduction: why general models are not enough

1.1 Where this started

The system described here did not start as a research programme. It started as a side effect of trading work. In the founder's words:

> "Was creating trading signals for equities, that's why I realized how GIFT Nifty and news affect indices, sectors, latency for an event."

Three things in that sentence shaped everything after it. News affects an index. It affects sectors differently from the index. And it does so with latency: the effect arrives later, and not everywhere at once.

The moment that turned an observation into a design came on a trip, which the founder dates to September 2024:

> "I was on a trip to Himachal where because of rains the village people told me that the yield was not enough, then I researched what happens when apples are not enough produced, that's when I came to realize India imports from Australia and New Zealand. The point was someone with knowledge of something like this gets enough time to order containers."

And on the year:

> "2024 September, it was a bad year for apples since unexpected rains did not give them good yield."

The record on disk begins later than this. The first commit is dated 14 March 2026, and nothing on the development machine carries a 2024 date. The origin therefore rests on the founder's account. What the record does show is that the case stayed with the system. When the first file of cause and effect rules entered version control on 26 April 2026, a rule for rain damage to Himachal apples was one of its first twenty-one entries (Appendix A, entry 4.3). The earlier draft of this paper used the same case as its example of stakeholder incidence.

The apple case contains the whole argument in small. A weather report from one state is, for most readers, a local story with a negative tone. For a grower it is lost income. For an importer it is an order to place. For a cold storage operator it is pricing power. For a state government it is relief spending. The information was public. What was scarce was the chain of reasoning from the rain to the container, and the time that chain buys.

1.2 The flip side

The second observation is that the chain never points one way:

> "We think any event always has one effect, but it's a flip of a coin, it has 2 sides. An event while beneficial for certain sectors becomes disadvantageous for other sectors."

Section 2 develops this formally. Here it is enough to say that it rules out the most common representation of news in finance, a single score from negative to positive. A score has one side.

1.3 The butterfly effect and the calendar

The third observation is about depth and about place:

> "Once I went down the rabbit hole, I started understanding the complexity of a single effect, the butterfly effect of it. And then how cultural context matters: during Diwali 15 days market slows down, near Chinese New Year 1 month supply chain gets disrupted, how Strait of Hormuz affected the entire world and energy costs."

A single event propagates through supply, credit, logistics and policy. And the same event lands differently depending on when it arrives. A shipment delayed in the weeks around Chinese New Year is not the same event as the same delay in another month. A person who has worked in a market for years carries this without thinking. A model trained on a general corpus carries it weakly, if at all.

1.4 Few events, large effects

The fourth observation makes the problem tractable:

> "Macro events are not many. Government taxation (GST) affects entire market. FTA and other agreements which are sub-events of macro events have net effect which are huge. Government schemes, geopolitics, they are all part of how the money moves, why FII exit hurts India."

If the events that move a national economy were unbounded, a curated layer would be hopeless. They are not. The live event taxonomy holds 162 events as of this draft (Appendix A, entry 8.5). The work is not in the count. It is in knowing, for each one, who gains, who loses, through which channel and after what delay, and in keeping that knowledge true as the country changes.

1.5 The compression thesis

What should such a system give back? The founder's answer:

> "It would be like I am summarizing 10 hours of critical thinking into 1 answer."

This is the product definition. The output is not a sentiment, a summary of the article, or a recommendation. It is the compressed result of the reasoning a careful analyst would do about one event: the event identified, the sectors that gain and lose, the people affected, the conditions, and how sure the system is. We call the thing that carries this a structured object, and Section 6 defines it.

1.6 The claim

The founder states the claim this way:

> "LLMs cannot be country agnostic, and we need an intelligence layer that has no training cutoff date, rather they are trained every day."

We read the first half as follows. A general model is built to be the same everywhere. That is its strength for language and its weakness for decisions, because a decision is always made somewhere. The home page of the project puts the same point in plainer terms:

> "A general-purpose model trained on aggregated English data treats India as one cultural bucket."

and, of the systems built on this view, "Each one exists because a general model failed at something specific."

The second half needs care, because this paper describes the system as it exists and not as intended. The layer has no training cutoff in this sense: its sources are read every day, headlines it cannot place are stored and tested again every day, new events are proposed every week, and its knowledge files change whenever a person approves a change. Its neural weights are not retrained daily. They were last retrained on 2 September 2026, and scheduled retraining has been switched off since 9 August 2026 (Appendix A, entries 6.11 and 7.2). What is current every day is the knowledge, not the weights. Section 3 argues that this is the right place for currency to live.

The thesis of the paper, then: general models are general, not contextual. A country-specific decision needs a country-specific layer between the model and the person, and that layer must be able to refuse, to learn from its refusals under human approval, and to be checked against what happened.

There is a name for this layer. We call this layer the bridge. A general model can read a sentence in any language. It cannot, by itself, cross from reading the sentence to knowing what the sentence means for one country or one sector. The bridge is what carries a general capability across that gap, from language to a particular place and a particular industry. It is the contextual layer named above, and the rest of this paper is an account of how one bridge was built, how it failed, and what a second one would require.

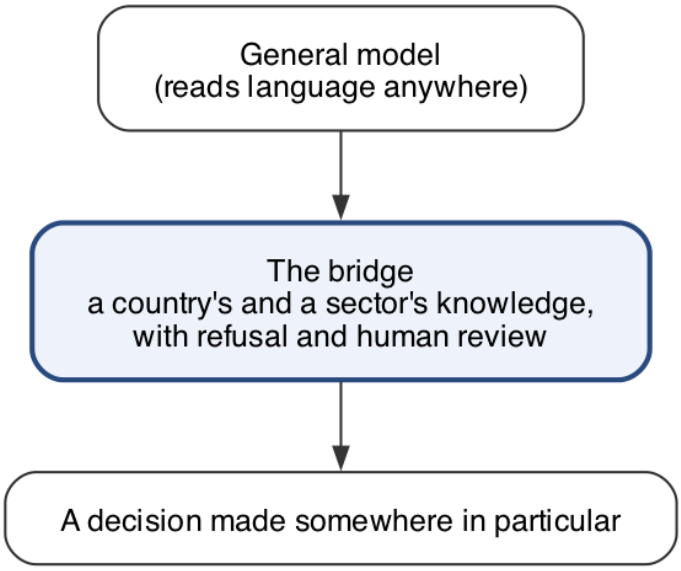


Figure 1. The bridge. A general model reaches a decision in a particular country and industry through the contextual layer.

2. Why scalar sentiment is the wrong unit

2.1 The usual framing

Traditional financial language processing maps a headline to a polarity: positive, negative or neutral. The framing is convenient and economically incomplete. A headline usually touches several balance sheets at once. One agent's input cost is another's revenue. One sector's subsidy is another's tax burden.

We use five terms throughout.

- Incidence: the distribution of gain, loss, burden or exposure across economic agents.
- Redistribution: a transfer of margin, cost, tax burden, liquidity or risk from one group to another.
- Transmission channel: the path by which an event reaches a sector or a stakeholder.
- Propagation: delayed or second-order movement through linked sectors.
- Exposure: a sector's sensitivity to a given pressure.

The object of interest is not sentiment. It is incidence: who is affected, through which channel, in which direction, after what delay, and in which market regime.

Formally, scalar sentiment projects an event graph onto one dimension:

```
phi: EventGraph -> R
```

The system described here maps it into sector space:

```
psi: EventGraph -> R^N
```

with `N = 32` in the current public schema. The point is not that a scalar is always wrong. It is that it is a lossy projection of a graph, and what it loses is the part a decision needs.

2.2 Events are redistributive graphs

The central statement of the earlier draft stands:

```
Market events are redistributive graphs, not scalar sentiment objects.
```

A redistributive graph has nodes (sectors, stakeholders, policy actors, suppliers, consumers, capital providers), edges (cost pass-through, credit tightening, subsidy transfer, tax incidence, inventory repricing, import exposure, logistics disruption), signs (who gains, who loses), lags (when each channel acts) and regime modifiers (whether volatility or liquidity changes the reading).

This is the founder's coin with two sides, written as a data structure. Many events redistribute value before they create or destroy any:

- A fall in crude oil moves margin from upstream producers to fuel users, aviation, paints, chemicals, logistics and the fiscal account.
- A repo rate hike moves burden toward leveraged borrowers and rate-sensitive sectors, while some deposit, currency and inflation-control channels improve.

- A crop failure moves value from growers and rural consumers toward substitute suppliers, importers, storage operators and processors holding inventory.
- A logistics disruption moves value from exporters and time-sensitive chains toward competitors, other routes and holders of stock.

The question is never "is this headline positive?" It is:

```
positive for whom, negative for whom, through which channel, and after what delay?
```

2.3 The record shows the failure

This is not only an argument. It is what the development record shows. The first model, committed on 14 March 2026, was a three-class sentiment classifier. Twelve days later the journal for the next release recorded what went wrong (Appendix A, entry 2.1):

> "The v2 model had ~50% labeling errors in training data"

with examples such as a headline about markets crashing predicted as positive. The cause was recorded in the same entry: labels had come from a general-purpose sentiment lexicon that did not understand financial context. The repair was not a better lexicon. It was a change of unit. Labels were taken from a list of market events, each with a known direction of effect, and the journal of that day records the first event taxonomy of 136 events.

Even then, the same journal lists as a known limitation that a central bank rate hike was still predicted as neutral, because its tone is factual. A rate hike has no tone. It has incidence.

Ten days after that, the changelog for the first multi-output model recorded the first explicit two-sided rule in the record (Appendix A, entry 3.2):

> "Context rules override polarity: e.g. rupee strengthens -> negative sentiment (hurts exporters)"

The word "strengthens" is positive in any lexicon. For an exporter paid in dollars it is a margin cut. The system had to be taught that the same word points two ways depending on who is reading.

2.4 A failure matrix

Headline type	What scalar sentiment says	What is redistributed
Rupee appreciation	A positive currency move	Importers and oil-sensitive sectors can gain; software and pharmaceutical exporters can lose margin.
Crude decline	Positive for the economy	Aviation, paints, chemicals, logistics and consumers may gain; upstream energy and some fiscal channels weaken.
Export ban	Positive for inflation control	Domestic consumers may gain; producers lose pricing

Headline type	What scalar sentiment says	What is redistributed
		power; competitors abroad gain share.
Flooding	Negative weather	Logistics, agriculture and insurers suffer; storage, repair, reconstruction and substitute suppliers may gain.
Repo hike	Negative for growth	Borrowers and rate-sensitive sectors are pressed; deposit channels, currency support and inflation control can improve.

2.5 Worked example: imports, the rupee and metals

The founder's account of the flip side continues:

> "This led me to understand the more we import, the more rupee devaluates, then dollar gets stronger and metals suffer the most in that case."

We set this out as a chain, because it shows first-order and second-order transmission in one case. It begins where the apple story ends.

```
A domestic shortfall (a poor harvest, a supply gap)
  -> more is imported
  -> more foreign currency is bought to pay for it
  -> pressure on the rupee
  -> a stronger dollar in rupee terms
  -> every dollar-priced input costs more
  -> sectors that buy dollar-priced inputs and cannot pass the cost on
lose margin
```

Read as incidence:

Step	Who is on the gaining side	Who is on the losing side
Shortfall at home	Substitute suppliers, importers, holders of stock	Growers, rural consumers
Larger import bill	Foreign suppliers, shippers, ports	The trade balance; the fiscal account if duties are cut
Weaker rupee	Exporters paid in dollars	Importers of fuel, fertilizer, electronics and industrial inputs
Dollar-priced inputs rise	Domestic producers of the same input, where they exist	Users of imported metals and ores; firms with dollar debt

Three cautions keep this honest. First, the chain is a structured hypothesis about transmission, not a measured law. Its links differ in strength and timing, and one harvest does not move a

currency by itself. The founder's point is about the direction in which repeated imports push. Second, the effect on metals is not one effect. A domestic producer of a metal may gain from a weaker rupee while a manufacturer that buys the metal loses, which is the flip side appearing again inside one sector label. The development record contains exactly this lesson: an internal probe in May 2026 found that a broad energy bucket had mixed upstream gainers with refiners under margin pressure, and that the fix was to split the sector, not to adjust a score (Section 9.3). Third, a scalar model has nowhere to put any of this. It would read the first headline as mildly negative weather news and stop.

The system could not represent this chain at all until September 2026. The internal changelog for 2 September 2026 records that "Import events had no representation at all", because the patterns that recognise a policy action covered exports only, and that import actions and import rules were added on that day (Appendix A, entry 7.6). The founder's insight predates, by his account, the system's ability to express it by about two years. That gap is part of the argument of this paper: the knowledge lives in a person first, and the layer is the slow work of writing it down.

A second case, larger and more recent, shows the same shape at the scale of a national staple. India is the world's largest exporter of rice and one of the largest producers of wheat, and in recent years it has restricted the export of the very grains it leads in. After a heatwave cut the 2022 wheat harvest and the government revised its production estimate down from about 111 to about 105 million tonnes, India banned most wheat exports in May 2022 to hold down domestic prices. In July 2023 it banned exports of non-basmati white rice for the same reason, a restriction it lifted only in September 2024 under a minimum export price. Each ban is a redistributive event with two sides: domestic consumers and the fiscal account are protected, while the country's own exporters lose sales, importing countries that depend on Indian grain face higher prices, and rival exporters elsewhere gain the share India gives up. The same country has also crossed the line it usually sells across. After a weak harvest, India was a net importer of wheat in 2016 and 2017, with private millers buying several million tonnes from abroad, and in 2024, with government wheat stocks at a sixteen-year low, it publicly weighed importing again before the government said its stocks were sufficient. A grain that India is known to export is, under domestic pressure, a grain India can be made to buy.

None of this required a forecast. Each turn was preceded by public, dated signals: the 2022 heatwave and the downward revision of the wheat crop before the export ban; the drawdown of government stocks and the rise in domestic prices before the rice restriction and before the 2024 import question. A structured record of the event and its chain, the kind this paper describes, would have produced the incidence at the moment those signals were public: who gains, who loses, through which channel, and after what delay. That is the same advantage the apple importer had in Section 1.1, written down as a data structure instead of carried in one person's head. It is not a prediction of a price. It is the compressed reading of an event whose parts were already on the record, available to anyone who had the chain and too easy to miss for anyone who did not. The public sources for the facts in this example are listed at the end of the paper.

2.6 Worked example: a change in the goods and services tax

The largest and most frequent macro events in India are not weather or trade. They are policy. A change in the goods and services tax, the budget, a free trade agreement or a government scheme moves value across the whole economy at once, which is why the founder's list of the events that

matter begins with taxation (Section 1.4). A tax change is a redistributive event in the plainest form. When the GST Council moved to a two-rate structure in September 2025, effective on 22 September, it cut the rate on a long list of mass-consumption goods, soaps, shampoo and toothpaste from 18 to 5 percent, televisions, air conditioners and dishwashers from 28 to 18 percent, and staples such as milk, paneer and Indian breads to zero, while moving sin and luxury goods to a new 40 percent rate. The two sides are immediate: makers and buyers of the goods whose rate fell gain, through lower prices and the demand that follows; producers of the goods whose rate rose are pressed; the fiscal account gives up revenue on one side and recovers it on the other; and the timing itself redistributes, because a buyer who hears of a coming cut will wait for it, moving demand across the date. A scalar reading of the headline as broadly positive records none of this.

The same event type can carry the opposite incidence. When the goods and services tax was first introduced on 1 July 2017, the event was not a price cut but a compliance shock: by one industry estimate about seventy percent of small and mid-sized firms had not begun to prepare, the filing portal buckled under load, and the burden fell hardest on small traders who had kept paper ledgers, on their working capital, and on exporters waiting for refunds. One is a demand event for consumers and consumer-facing sectors; the other is a disruption event for small enterprises and compliance-heavy chains. This is the paper's argument at the level of the event class that matters most for India. Policy is not one layer among many; it is the largest source of redistributive events, and it is where a country-specific layer earns its place, because the incidence of a tax change runs through that country's sector structure, its informal economy and its compliance base, none of which a general model holds. A structured record reads the policy notification and returns which sectors gain, which are pressed, through which channel and after what delay. The public sources for the dates and the rate changes are listed at the end of the paper.

2.7 First order and second order

First-order incidence is direct: a flood that damages roads cuts transport capacity. Second-order incidence is propagated: lost transport raises spoilage, which raises prices, which draws relief spending, changes procurement and weakens credit-sensitive consumption. The framework keeps three things apart: direct incidence, downstream incidence, and the net sector vector after both are combined.

2.8 Why a country matters

The same text does not carry the same incidence in two economies:

```
Incidence(event, economy_a) != Incidence(event, economy_b)
```

A fuel price change in a high-income economy with little subsidy does not have the incidence it has where household budgets, freight costs, excise collections, fuel retailer margins and rural purchasing power are tightly linked. The same holds for food inflation, fertilizer policy, rainfall, import duties, export bans and currency moves. Incidence depends on subsidy structure, household composition, import dependence, the share of the informal economy, agricultural sensitivity and how policy is transmitted. Section 8 takes this further: it holds inside a country too.

2.9 Foundations in the literature

The framework rests on established work, which the earlier draft set out in detail and which Appendix D lists in full. That sectors are linked, so that a shock cannot be judged in isolation, goes back to Leontief (1936) and Hirschman (1958); that local shocks become aggregate ones through network structure is the result of Gabaix (2011) and Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi (2012); and Inoue and Todo (2024) show news about one firm reaching its suppliers and customers. Bernanke, Gertler and Gilchrist (1999) describe how credit amplifies a disturbance, Kilian (2009) shows that the same oil price move can carry different causes, and Whaley (1993) motivates reading events by volatility regime. On method, Shannon (1948) underlies the use of token information against single-keyword collapse, Granger (1969) the cautious use of lagged predictability to find candidate rules, and Pearl (2000) the separation of association from structured causal reasoning. In language processing, Chen and co-authors (2019) show the value of fine-grained events, Araci (2019) provides the polarity baseline that this paper argues is a different task, Overbeck and co-authors (2025) argue that sentiment must be tied to the object evaluated, and Lyu and co-authors (2022) give a precedent for extracting financial causality from text. These references motivate the framework. They do not imply that the system reproduces any one of their models.

3. The three-part stack

The system is best understood as the middle of three parts. None of the three is sufficient alone.

3.1 The general model

A general language model supplies language and reasoning. It can read a headline in any phrasing, explain a chain of effects in fluent prose, and answer follow-up questions. For the task in this paper it has four limits.

- It has no country. Its beliefs about an economy are an average over whatever text it saw.
- It has a date. Its knowledge stops at some point and it cannot reliably tell which of its beliefs are stale.
- It does not refuse well. Asked about an event it does not understand, it tends to answer anyway.
- It leaves no audit trail. Two identical questions can produce different chains of reasoning, and neither can be traced to a rule that a person approved.

None of these is a criticism of general models as such. They are what general means. In this stack the general model is the reader and the speaker. It is not the source of the facts.

3.2 The contextual layer

The contextual layer is the subject of this paper. As built, it consists of:

- an event taxonomy: the list of events that matter in this country, each with its direction and sector exposure;
- a rule layer: curated cause and effect rules for chains that a list of events cannot hold, such as weather to crop to price to policy;

- primitive pressures: a small set of economic forces through which events act on sectors;
- propagation: delayed movement through linked sectors;
- a context library: the calendar, seasons, fiscal dates and standing geopolitical conditions of the country and of the countries it depends on;
- a refusal branch: an explicit outcome for "this cannot be placed", which is stored, not hidden;
- a review loop: refusals become proposals, and proposals become knowledge only when a person approves them;
- an outcome monitor: every claim that an event moves a sector is compared with what the sector did.

The layer is deterministic first. Evidence found in the text decides before any neural model is consulted, and a model's answer is accepted only when nothing better is available and the model itself is not abstaining. This ordering was not the original design. It was forced by a failure, described in Section 5.

The layer does not write prose, hold a conversation, or give advice. It produces one structured object per event.

This layer is the bridge named in Section 1.6. It carries a general capability across one gap: from reading a headline to knowing, for this country, whom the headline moves and in which direction.

3.3 The human

The third part is a person. In this system a person does three things that neither of the other parts can do.

- Approves growth. No event enters the taxonomy without a human decision. Proposals are drafted automatically each week; automatic promotion is off by default (Appendix A, entry 7.4).
- Reviews refusals. The stored refusals are the system's statement of what it does not know. A person decides which of them reveal a missing event, which are noise, and which are events that are real and move nothing.
- Supplies judgment about anchors. A person decides when the layer does not have enough information to justify a magnitude. When a foreign tariff event was split by direction, one variant was held back because no honest anchor existed for its magnitude. A model can propose a value. Only a person can decide that there is not yet a basis for one.

3.4 Why each part is necessary and none is sufficient

Without	What fails	Evidence in this record
The contextual layer	The general model answers fluently from an average of the world, with no country, no date and no refusal.	The first models here were general-purpose in spirit and mislabelled about half their training data (entry 2.1).
The general model	The layer reads only	On the day of a central bank

Without	What fails	Evidence in this record
	phrasings it has been taught and cannot converse or explain.	rate hike in October 2026, plainly worded announcements of the hike were refused while other phrasings of the same event were placed (Section 10).
The human	The layer learns from its own output and drifts.	The retrained model of September 2026 is described in the record as "a distillation of the rule system rather than an independent signal" (entry 7.7). An unattended loop re-queued the same headlines indefinitely (entry 6.11).

The stack did not emerge fully formed. Section 5 shows how each part was added in response to a specific failure.

4. The architecture as built

This section describes the system as it exists on the date of this draft. It keeps the earlier draft's description of the reasoning pipeline and adds three components that the earlier draft treated as surroundings: ingestion, the review loop and the outcome monitor. They are first-class parts of the system.

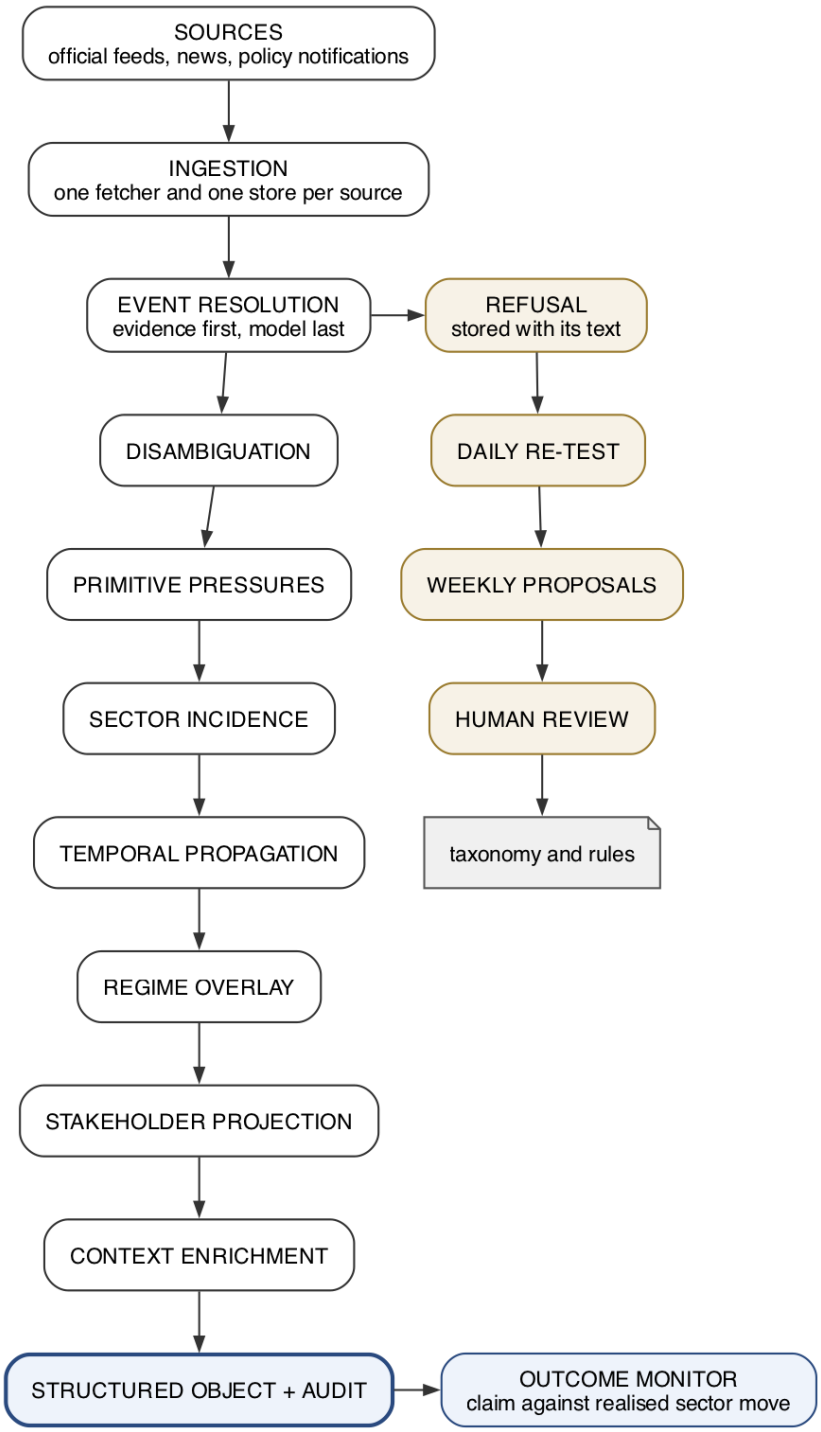


Figure 2. The architecture as built. The main pipeline runs from sources to the structured object; event resolution branches to a stored refusal that feeds the daily re-test, the weekly proposals and human review, and the structured object feeds the outcome monitor.

4.1 Ingestion

The first version of the system analysed whatever text it was handed. The present system reads its own sources. The shared store records thirteen labelled sources as of this draft: exchange filings and circulars, central bank and government press releases, regulator orders, four news sources, and three policy sources covering trade notifications, indirect tax notifications and commerce ministry releases (Appendix A, entries 5.2, 6.9 and 8.3).

Ingestion is governed by five rules adopted on 8 October 2026, recorded as the founder's wording: one fetcher per external source, one store per source, one client per external host, an alert on any run that stores nothing, and a source label on every row (entry 8.4). The rules were written after an audit found, among other things, a feed that had stored nothing for nine months while its job reported success.

Two properties of ingestion matter for the argument. It is what removes the training cutoff: the layer's view of the country is as recent as its last fetch. And it supplies the article body as well as the headline. Until 9 August 2026 the ingesters collected article bodies and the classifier never received them (entry 6.10).

4.2 Event resolution

Event resolution identifies the market-relevant event in a text. It is an arbitration among several sources of evidence, in a fixed order of trust:

1. a rule recognised in the text by a classifier trained only for that purpose;
2. a deterministic cause and effect rule matched in the text;
3. a taxonomy event matched by its vocabulary;
4. the event output of the IMI multi-task model, whose task-specific heads we train on an open-source transformer base encoder, accepted only when it is confident, is not abstaining, and no rule evidence contradicts it;
5. otherwise, refusal.

The principle is recorded in the system's own status document as "Evidence outranks inference." The numeric conditions at each step are not published. After arbitration, a guard can still turn a weak answer on a text with no financial content into a refusal.

4.3 Refusal

Refusal is an output, not an error. When no step claims the event, the system returns no event and no sector vector, and stores the text with its source, the candidates it considered and the time. The store has existed since 9 May 2026 (entry 5.1). It is the most important table in the system, because it is the system's own record of what it does not know.

Refusals have several terminal states, each of which means something different: placed after a later re-test; closed because the text is not financial; closed because the event is real and recognised but moves nothing; approved by a person as a new or corrected event; or sent to a person because repeated re-tests did not settle it. The third state was added for a specific reason. A daily record of yesterday's money market operations is correctly identified and legitimately

moves no sector. The design note that introduced the state puts it in one line: the system "classifies events, not states" (entry 6.12).

4.4 Disambiguation

Financial text is full of overloaded words. "Crash" can describe an index, a commodity, an aircraft, a server or earnings. The system uses the information each token carries about each event, and the tendency of events to follow one another, when the primary evidence is uncertain. The memory behind this is built from captured events and from human-approved refusals, so an approved decision improves later disambiguation (entry 6.7). The aim is not to have a model explain the headline. It is to stop a salient word from hijacking the answer.

4.5 Primitive pressures

Events act on sectors through a small set of economic forces. The current public set has eleven:

Primitive	Economic role
`supply_shock`	Availability of goods or services.
`demand_shock`	Consumption or investment demand.
`cost_push`	Input-cost pressure.
`liquidity`	System liquidity.
`credit_availability`	Ease of borrowing.
`policy_restriction`	Taxes, bans, quotas or regulatory constraints.
`export_competitiveness`	Export advantage or disadvantage.
`import_dependency`	Exposure to imported inputs.
`logistics_disruption`	Transport or storage constraints.
`risk_sentiment`	Risk appetite and risk-off pressure.
`inflation_expectation`	Forward inflation pressure.

Primitives prevent an explosion of rules. Instead of mapping every event to every sector, the system maps events to pressures and pressures to sector exposure. The primitive definitions and a matrix from pressures to sectors first entered version control on 26 April 2026 (entry 4.2).

4.6 Sector incidence

The sector step turns pressures and event exposure into a signed value for each of 32 sectors. It is the step that keeps winners and losers together in one answer:

```
Event: crude oil price falls
Pressures:
  cost push falls
  import dependency eases for importers
  upstream revenue pressure rises
Sector incidence:
  Aviation: favourable
  Paints and chemicals: favourable
```

```
  Logistics: favourable
  Upstream energy: adverse
```

The example is structural. No production coefficient is published in this paper.

4.7 Temporal propagation

Propagation carries second-order effects forward in time:

```
Crop loss
  -> food inflation
  -> risk of policy tightening
  -> pressure on credit availability
  -> automobiles, real estate and discretionary consumption
```

A headline can carry a delayed path even when the sector first affected is obvious. Rules carry their delays explicitly, at fixed horizons.

4.8 Regime overlay

The volatility regime changes how a sector vector should be read. The same vector in a calm market and in a panic is not the same information. The overlay scales the vector by regime. It is useful and incomplete: it does not yet take wider liquidity conditions into account.

4.9 Stakeholder projection

The stakeholder step projects the sector vector onto roles: producer, trader or intermediary, investor, government or fiscal authority, and international trade actor. For the founding case, "unseasonal rain damages Himachal apple orchards":

- growers lose crop value;
- substitute suppliers and importers may gain;
- cold storage operators may gain pricing power;
- the government faces relief spending and inflation monitoring;
- investors look to listed storage, consumer goods, logistics and agriculture-linked companies.

The stakeholder views are derived from the structured output. They are not free narrative. A stakeholder mapper first entered version control on 3 May 2026 (entry 4.4).

4.10 Context enrichment

The last step consults a deterministic library of calendar, seasonal, fiscal, institutional and geopolitical conditions and appends what it finds. This is the founder's fourth observation built as a component. The library is organised by kind of phenomenon, with the region as a label, and its entries for the festival season, for the factory shutdown around Chinese New Year and for the Strait of Hormuz are the cases named in his account. The design note states what the component may never do: retrain itself, edit the taxonomy, or change the classifier's output. It only appends.

Two facts about its present state should be stated plainly. Until 19 September 2026 a failure inside this component was swallowed silently, so a broken library produced answers with no context and no trace (entry 7.8). And since 21 September 2026 its output is not included in the public response (entry 7.10). It runs on every request; what the public contract returns does not yet include it.

4.11 The review loop

The review loop turns refusals into knowledge under human control. It has four stages.

1. Daily re-test. Every stored refusal is analysed again with the current engine. Many refusals turn out to be the engine's past mistakes, not gaps in knowledge.
2. Weekly proposals. Repeated refusals that share a pattern are drafted as proposed events, with a suggested sector exposure borrowed from the nearest existing event and labelled as borrowed.
3. Human review. A person approves, edits or rejects. Approval writes the event into the live taxonomy, updates a hashed manifest of the knowledge files, and reloads the runtime.
4. Memory. The approved decision joins the history used for disambiguation.

Automatic promotion exists as a mechanism and is off by default. As of this draft the queue holds more refusals awaiting a person than one person can clear, which Section 10 treats as a limit.

4.12 The outcome monitor

Every claim of the form "this event moves this sector, in this direction, at this delay" is a hypothesis that the market can contradict. A daily job compares each such claim with realised sector returns adjusted for the broad market, at several delays, and flags the claims whose record is weak. Flagged claims go to the same human queue as proposed events. The monitor's own results are provisional, for reasons given in Section 9.

4.13 The public contract

The public output has not changed in shape since the earlier draft:

```
{
  "headline": "string",
  "event": "string|null",
  "confidence": "float",
  "vix_regime": "string",
  "sector_vector": {},
  "top_positive_sectors": {},
  "top_negative_sectors": {},
  "stakeholder_views": {},
  "raw_assignment": {
    "resolved_event_id": "string|null",
    "cause_effect_rule_id": "string|null",
    "weather_triggered": "bool"
```

```
  }
}
```

`sector_vector` is the canonical key. An `event` of null is a refusal. Older wording such as `sector_impacts` or any execution-like language is not part of the contract.

4.14 A worked example

For the headline "RBI hikes repo rate by 25 bps", the pipeline produces, in outline:

Stage	Output
Event resolution	A monetary tightening event, resolved by taxonomy evidence.
Disambiguation	"repo rate" and the central bank's name resolve to monetary policy, not general banking news.
Pressures	Liquidity, credit availability, policy restriction and inflation expectation.
Regime	Read from the volatility index on the day.
Sector incidence	Adverse for real estate, banks, non-bank lenders, financial services and automobiles; milder for consumer goods; favourable on the exporter side for software services.
Stakeholder views	Investor, fiscal and trade views list different sectors, because a listed-sector exposure is not the same as a fiscal or a trade exposure.
Audit	The resolved event, whether a fallback was used, whether a weather rule fired.

Exact values are omitted. The point of the example is that a rate hike is not "negative". It redistributes: borrowing is pressed, some currency-linked channels differ, and the three stakeholder views are not copies of one another. Section 10 returns to this same event to show a present weakness.

5. The development arc: static to dynamic

The record is treated here as scientific evidence: a log of assumptions, the failures that contradicted them, and the changes that followed. Every date below is taken from the timeline in Appendix A, which gives the file and line for each. Where the record states what changed and not why, we say so and do not supply a reason.

A note on the boundaries of the record. It begins on 14 March 2026. The two years before that, from the Himachal trip through the trading signal work, are known only from the founder's account. And from 5 June 2026 the engine was deliberately removed from version control, so the later record is changelogs, audits, backups and database rows.

5.1 Stage one: a sentiment model (March 2026)

The first commit, on 14 March 2026, is a market sentiment engine: a three-class classifier trained on stored headlines, with a ticker to sector map, a volatility adjustment and a news impact lookup around it. The journal of that day describes it as tuned for Indian markets and for intraday trading. It reports very high validation accuracy.

The assumption was that a headline has a polarity and that polarity is useful. The failure arrived within twelve days. On 25 March the public model was withdrawn with the notice "A data labeling inconsistency was discovered in the training dataset." The journal of 26 March gives the finding: about half the training labels were wrong, because they came from a general lexicon with no financial context.

Two lessons are in that fortnight. Validation accuracy measured agreement with the labels, and the labels were wrong, so a number above 98 percent described nothing real. And the commit history shows the claims being withdrawn: on 15 March trading performance figures were removed from the description, and on 25 April a commit titled "Remove unsupported accuracy claims" deleted 385 lines of the public page.

5.2 Stage two: events (26 March 2026)

The repair changed the unit of analysis. The journal:

> "Created 136 market events with known sentiment"

and each event carried a base impact at three levels of severity. Labels were now derived from which event a headline matched. This is the first move from sentiment to event classification, and the first taxonomy file in version control dates from the same commit.

The failure that remained was recorded the same day: only about 40 percent of headlines matched any event, the model "often returns 100% confidence scores", and a rate hike still read as neutral. An event list repaired the labels. It did not yet produce incidence, because one event still had one sign.

5.3 Stage three: one model, four outputs (April 2026)

On 5 April 2026 a multi-task model replaced the classifier, with outputs for sentiment, event, a macro signal and a vector across 32 sectors. This is the first sector vector. The same changelog entry records the first rule that reverses polarity by reader, for currency moves (Section 2.3), and the first connection to a live source: the pre-market news crawler began publishing each item with its event, macro signal and sector vector.

The failures of this stage are recorded in the following days. On 6 April severe negative events were still being read as positive, and the response was an override at inference time plus four thousand synthetic training examples. The earlier draft of this paper drew the lesson that synthetic data can patch a known gap and cannot stand in for the real distribution. On 25 April experiments on real headlines made the point with a number: specific rules matched 165 of 72,974 real headlines. Rules were not abandoned. They were repositioned as high-precision anchors inside a wider resolver.

On 12 April a further component computed the sensitivity of each sector to four market factors from market data on a rolling window. The changelog entry calls it "Layer 4". It is the first appearance of the word layer for this system, and the first time anything in it was tied to realised market data instead of to authored values.

5.4 Stage four: rules, primitives, stakeholders (26 April to 3 May 2026)

In one commit on 26 April 2026 three things entered version control together: a file of twenty-one curated cause and effect rules "for weather, trade, shipping, fiscal, insurance, and competitor cascades", the definitions of the primitive pressures, and a matrix from pressures to sectors. One week later, on 3 May, a stakeholder mapper and a propagation engine were committed alongside the first hosted interface.

The record does not explain why these arrived when they did. There is no changelog entry for the 26 April commit that states a reason, and the rules file is named version 3 at its first appearance, with no earlier version on disk. We mark this as unknown (Appendix A, entries 4.1, 4.2 and 4.4). What can be said is what the content shows: the rules encode chains with delays, which an event list cannot hold, and the Himachal apple case is among them.

On the same day as the stakeholder mapper, the public output was standardised on the sector vector and all long and short wording was removed. The earlier draft records the reason: execution-like language blurred structured evidence with trade instruction.

5.5 Stage five: refusal and fallback (May 2026)

The first stored refusal is dated 9 May 2026. From that day, a headline the system could not place was written to a table instead of being dropped. The first official sources were connected on the same day.

No decision record for the refusal store survives on disk, and we do not invent one. Its effect is clear from what followed. Two days later a dynamic fallback was added for headlines with no specific rule, with the explicit provision that its output be marked as telemetry and logged for review. The earlier draft summarised the correction: fallback became telemetry, not authority. The same day the sector polarity of a rate hike, the event that had read as neutral on 26 March, was corrected to be conditional on the sector.

The first review loop dates from the same week. An audit written in July reconstructs it: a weekly script gathered the refusals, suggested rules and sent a digest; a person edited a file of accepted rules; a gated monthly job retrained. On 15 May a changelog entry describes closing "automation gaps across the self-evolving model loop", so that the weekly review could generate training data and trigger a fine-tune behind a gate. This is the first scheduled retraining path.

This is the first pivot. Before 9 May the system had two outcomes, an answer or nothing. After it, the system had a third: a recorded statement that it did not know. Everything dynamic in the present system grows from that table.

5.6 Stage six: the loop closes, then breaks (June to August 2026)

On 14 June 2026 the review loop was rebuilt as an authenticated review page. The changelog states what approval does:

> "implemented approval flow that upserts reviewed events into ... `taxonomy_india_v2_calibrated.yaml`, updates `taxonomy_manifesto.json`, reloads IMI runtime, and marks matching null events as `approved`"

This is the first closure of the loop in the full sense: a refusal, reviewed by a person, becomes a change in the live knowledge without a release. The July audit is careful about what this was and was not. It was "manual->hybrid (dashboard + suggestion), never fully automatic". The machine suggests; a person promotes.

Then the system spent two months failing in ways that only a running loop can fail. The record is specific about each.

- A migration broke the feeds. The engine was relocated in mid-June behind an authorisation gate that the internal feed jobs were never given credentials for. The repair report of 16 July concludes: "This was one incomplete migration, not three independent regressions."
- A cache write destroyed the taxonomy. Between 26 June and 1 July a small file of a different kind was written over the cached taxonomy. Every analysis failed, and the pre-open predictor that depends on the layer missed its morning window. Guards were added on the read side and on both write sides.
- The backlog was misread. By 16 July there were 1,883 refusals awaiting review. The analysis written that day reframed the problem: "this is not 'approve 1,883 items.' It is 'understand why 1,883 items needed approval.'" Its finding was that 46 percent of them had been labelled correctly and had no sector mapping. They were not mysteries. They were one missing entry per event type.
- The matcher matched inside words. On 9 August an investigation found that a short keyword for a cricket league was matching inside "multiple", "principle" and "discipline", and returning that event at high confidence for unrelated corporate and regulatory headlines. The same defect affected other short keywords.
- The body never arrived. The same investigation found that ingesters had been collecting article bodies that the classifier never received.
- Approved events came back. Events promoted through review could produce a vector of zeros, which the system counted as a refusal, so the same headline returned to the queue indefinitely. A later entry, for a related loop found in September, counts one exchange circular queued 74 times.
- The scheduled jobs were failing silently. The weekly and fine-tune jobs had been failing without notice. Both were disabled on 9 August and an audit of job exit codes was added.

The response to the last four, on 9 August, was a daily job that re-tests every stored refusal against the corrected engine and records an explicit outcome for each. Its design note states the finding that justified it: "A large share of the backlog was therefore never genuinely unclassifiable - it was misjudged by a broken engine."

The lesson of this stage is the reason the third part of the stack exists. A loop that learns from refusals will also learn from the engine's own defects unless something outside the loop checks it. Here that something was a person reading the backlog as evidence about the system instead of as a list of chores.

The context layer was built in the middle of this period, in mid-July. The record gives its design and not the day of the decision.

5.7 Stage seven: evidence outranks inference (September 2026)

The second pivot is dated 2 September 2026. The internal changelog entry of that day begins with an admission:

> "The multi-task model's event label space had been contaminated since whatever labelled its training data ran the substring matcher ... Fixing the matcher never fixed the weights. This retrains them."

The defect of 9 August had been repaired in the matcher. It had not been repaired in the model, because the model had been trained on labels the defective matcher produced. The entry records five changes that together define the present system.

- The order of trust was reversed. "The classification cascade let a model guess outrank a taxonomy match. Rule evidence now runs first and completely."
- Refusal entered the model. The event output was retrained on labels from the corrected engine "with an explicit `no_match` class". The model can now abstain.
- Growth was scheduled and gated. A weekly job "drafts event proposals from the null-event backlog into a review queue. Auto-promotion is off by default."
- Outcomes were monitored. A daily job "validates every event-to-sector claim against realised, market-adjusted returns."
- Imports became representable (Section 2.5).

The same entry records two findings that no test had caught. A sign convention had been read two ways: an authored flag meaning "bad for this sector" was multiplied by a macro signal that was already negative for a bad event, and the two cancelled. The entry counts "85 sector entries across 46 events" that "computed a sign opposite to their own rationale". And about a third of the causal matrix had never reached the output, because sectors missing from one lookup table were skipped without notice. Both had been present while validation figures looked healthy.

The entry closes with a statement of the model's limit that this paper adopts as its own: the retrained model "is a distillation of the rule system rather than an independent signal". It learned from the engine's labels. Its agreement with the engine is not evidence that the engine is right.

Later in September two more silent failures were found by the same method. The context layer had been swallowing its own errors. And the log line that was supposed to record every unplaced headline had never once been written in production, because of how logging was configured. The stated purpose of that line, that missing coverage "shows up as data", had not been met for months.

5.8 Stage eight: policy sources and governed growth (October 2026)

On 5 October 2026 three policy sources were connected, the taxonomy grew by five events through the review route, and nine new policy action labels went live. The decisions recorded that day are small and characteristic. Foreign tariffs were split into separate events by direction "because a modifier cannot change sign". One variant was held back until there is "an honest

anchor" for its magnitude. One label was deliberately not added because it "would switch on a rule that misreads duties imposed abroad". A family of tax events was not promoted for the same reason of having no honest anchor.

On 8 October the ingestion rules of Section 4.1 were adopted. With that, the transition from a static model to a dynamic system is complete in outline: the layer reads its own sources every day under stated rules, stores what it cannot place, re-tests it, proposes growth, waits for a person, and is checked against outcomes.

5.9 How the stack emerged

The three-part stack of Section 3 was not designed. Each part was added after the absence of that part caused a recorded failure.

Part	Added	The failure that forced it
Event taxonomy	26 March 2026	Half the sentiment labels were wrong.
Two-sided sector output	5 April 2026	"Strengthens" read as positive for exporters.
Rules as anchors	26 April 2026	The reason is not recorded. Experiments of the day before showed specific rules covering a fraction of a percent of real headlines.
Stakeholder views	3 May 2026	The reason is not recorded.
Refusal store	9 May 2026	The reason is not recorded. What it enabled is.
Fallback as telemetry	11 May 2026	Many headlines had no rule and no vector; fallback risked becoming authority.
Human review that writes knowledge	14 June 2026	The first loop edited a side file that was not the source of truth.
Integrity guards	July 2026	A small file overwrote the taxonomy and every analysis failed.
Daily re-test of refusals	9 August 2026	The backlog was largely the engine's own past mistakes.
Evidence before inference	2 September 2026	A model guess could outrank a taxonomy match, and the model had been trained on defective labels.
Outcome monitor	2 September 2026	Authored signs had been wrong for months with

Part	Added	The failure that forced it
		nothing to contradict them.
Ingestion rules	8 October 2026	A feed stored nothing for nine months while reporting success.

The pattern is the argument. A general model failed at context, so a contextual layer was built. The contextual layer failed at knowing its own limits, so refusal was built. Refusal without review accumulated, so a human loop was built. A human loop without outside evidence drifted, so an outcome monitor was built. Every part is there because the parts before it were not enough.

6. The structured object

6.1 The formal framework

Let a headline be `H`. Event resolution gives an event class or a refusal:

```
e = f_event(H),   e in Events or e = none
```

When `e = none` the computation stops and the refusal is stored. Otherwise the event activates primitive pressures:

```
P(e) in R^K,   K = 11
```

A sector response matrix maps pressures to direct sector incidence:

```
S_0 = W * P(e),   W in R^(N x K),   N = 32
```

Propagation carries incidence through a directed graph `G` of lagged sector links:

```
S_total = S_0 + sum_{t=1..T} lambda_t * G^t * S_0
```

which separates direct incidence `S_0` from higher-order incidence `G^t * S_0`. Stakeholder views are projections from sector space, one operator per role:

```
V_r = M_r * S_total
```

The regime adjusts the vector:

```
S_regime = R_z * S_total
```

Context enrichment appends a set `C` of matched calendar and standing conditions without changing anything before it. The output is:

```
Y = {e, P(e), S_regime, V, C, A}
```

where `A` is audit data: confidence, the rule or path that decided, fallback markers, and the versions of the knowledge files in force.

The notation is deliberately simple. It does not imply that every coefficient is known. It says that event classification, primitive decomposition, sector incidence, propagation, stakeholder projection and regime adjustment are one computation and not unrelated heuristics. In the

running system not every path passes through every operator: a taxonomy event can supply its sector exposure directly, and a rule can supply a timeline. The framework is the common shape.

6.2 What causal means here

In this paper causal means a structured transmission hypothesis:

```
event -> channel -> exposure -> incidence
```

It does not mean proven intervention-level causality, guaranteed price causation, or a claim that the graph is complete. The claim is representational. The system structures incidence and exposure. It does not claim that every vector becomes a realised price move.

6.3 The object is the product

The object `Y` is the product. Everything a reader sees is a rendering of it, made downstream.

- The object is complete without prose. Every field is typed and bounded. A program can use it without parsing language.
- The object is auditable. `A` names the path that produced the answer. Two identical inputs under the same knowledge files produce the same object.
- The object can be empty. A refusal is a valid object with no event. A reader that receives it knows that the layer did not know, which is information.
- The renderer cannot add facts. A general model asked to turn the object into a paragraph may choose words. It may not introduce a sector, a sign or a stakeholder that the object does not contain. The earlier draft listed "constrained briefing generation that only verbalizes structured outputs" as future work. It is still future work. The principle is fixed in advance.

The division of labour follows from Section 3. The layer decides what is true enough to say. The general model decides how to say it to this reader. The human decides what the layer is allowed to know.

7. The multi-output principle: one object, many readers

7.1 The intended readers

The founder did not build this for traders alone:

> "I expected SaaS builders to adapt this into their products, because it gives them real time analysis, and the purchase or sales heads can revise plans and strategies accordingly."

The sentence names two kinds of reader. The first is a builder who puts the object inside another product. The second is the person at the end of that product, a head of purchasing or sales, who does not want an object at all. That person wants to know whether to change an order. The same object must serve both, and the readers between them.

7.2 What each reader takes from the object

Appendix C gives the field-level mapping.

Reader	What they want	What they take from the object
Developer	A stable, typed payload to build on	The whole object, unchanged
Dashboard user	What happened and who it touches, at a glance	Event, top gaining and losing sectors, regime
A forecasting model	Compressed overnight information as features	Confidence and macro signal, aggregated over recent items
Subscriber to an alert channel	One line when something material happens, and silence otherwise	Event, the two or three largest sector moves, and the refusal state to decide silence
Chief financial officer	A two-page view of exposure by channel	Sector vector grouped by pressure; fiscal and trade stakeholder views; delays
Board	A weekly account of what changed and what it means	The week's events ranked by breadth of incidence; outcomes from the monitor
Compliance or regulatory author	A note on why a disclosure or filing may be affected	Event, audit path, policy source of the item
Purchase manager	Whether to bring an order forward, hold it, or change supplier	Import dependency and logistics pressures; delays; the context library's calendar entries

7.3 Why each reading is different

A developer wants nothing removed. Any summarising done upstream is a decision taken away from them. The contract is therefore the full object, and its stability across versions matters more to this reader than any single field.

A dashboard user wants rank, not magnitude. The top gaining and losing sectors exist as separate fields for this reason. They are a rendering decision made once, inside the object, so that every dashboard shows the same thing.

A forecasting model wants numbers that are comparable from day to day. It takes the confidence and the macro signal and ignores the rest. The earlier draft noted that this is a coarse use, and that incidence-aware features such as the breadth of gaining and losing sectors would use more of the object.

An alert subscriber wants silence most of the time. For this reader the refusal branch and the "recognised, moves nothing" state are the most valuable parts of the object, because they are what allows a renderer to send nothing.

A chief financial officer reads by channel, not by event. The question is not what happened but through which line of the accounts it arrives: input cost, borrowing cost, currency, tax. The primitive pressures are that reading. They exist in the object so that a briefing can be organised around them.

A chief executive reads the same object for a different decision. Where the chief financial officer reads by channel, whether to hedge, when to commit capital expenditure, how input costs, credit lines and working capital move, and those are the primitive pressures read as lines of the accounts, the chief executive reads by position: strategic exposure, a competitor's likely move, which part of the business sits in the path of the event. Those are the stakeholder views and the propagation timeline, the first naming who is affected and in what role, the second naming in what order and after what delay. The object serves both because it carries the pressures, the stakeholder views and the timeline at once, and each officer takes the fields their decision is made of. This is the product thesis at the scale of a single reader: not a feed of headlines, but the compressed incidence of one event, addressed by the field that the reader's decision actually consumes.

A board wants the week, and wants to know whether last week's reading was right. This is the one reader for whom the outcome monitor is part of the rendering.

A compliance author wants the path. The audit data and the source label on the ingested item are the content; the sector vector is secondary.

A purchase manager is the reader in the founder's sentence and the reader in the apple story. This reader wants time. The fields that matter are the delays on each channel and the calendar entries from the context library, because the decision is when to order. The importer who heard about the rain in Himachal did not need a sentiment. They needed to know that a shortfall was coming, roughly when it would reach prices, and that containers take weeks.

7.4 Written once

The object is written once. The renderings are many, and none of them goes back to the text. This has three practical effects.

- Consistency. Every reader of the same event is looking at the same sectors with the same signs. A dashboard and a briefing cannot disagree about who gains.
- Cost. The expensive reasoning, the "10 hours", is done once per event and not once per reader.
- Control. A correction made in the taxonomy by one human decision changes every rendering at once.

This is why the system is not a sentiment engine with extra fields. A sentiment engine produces a judgment for one kind of reader. This system produces a compressed account of economic incidence, and leaves the judgment to whoever is reading. It is a compression layer for incidence.

7.5 Plugging the object into an existing workflow

A layer of this kind is not consumed directly. It is consumed through the object it produces. Four integration patterns are possible:

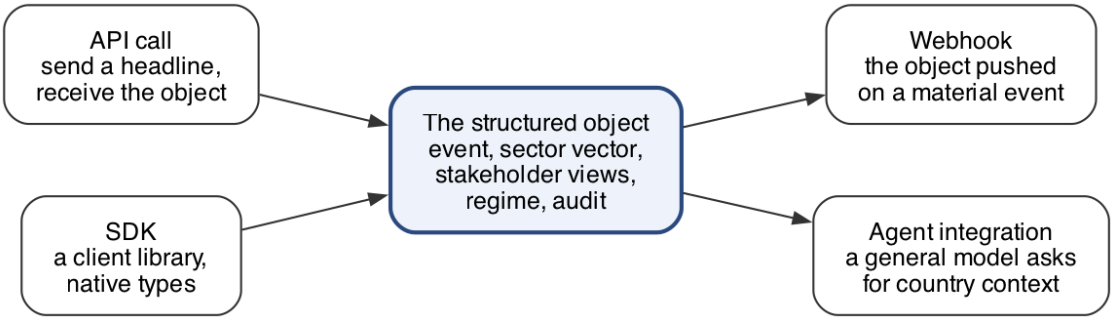


Figure 3. Four ways a builder integrates the object: an API call, an SDK, a webhook push, and agent integration. The object is the integration surface.

API call. The builder sends a headline, or an identifier for an ingested item, to the hosted interface and receives the structured object in reply. Nothing is installed. The builder's product holds no model and no knowledge files; it holds a request and a typed response, and it is insulated from every change inside the layer except a change to the object's contract. This is the pattern in use today for the developer reader (Appendix C).

SDK. The builder imports a client library into their own code. The library wraps the same interface and turns the object into the types of the builder's language, so that where the API pattern gives a request and a response, the SDK gives a function call and a native value. The builder receives the same object; what changes is that it arrives as something they can pass around their program rather than a payload they must parse. The client package exists today.

Webhook. The builder registers an endpoint, and the layer pushes an object to it when a material event is placed, rather than waiting to be asked. The builder receives, without polling, the event and its incidence at the moment the layer finishes resolving it. This is the pattern an alert product is built on, because it lets the builder stay silent until something moves. The push follows directly from the object; the rendering that decides what counts as material, and what counts as silence, is specified in Appendix C and is not yet built.

Agent integration. An autonomous agent, built on a general model, queries the layer for country context before it answers. The agent receives a structured object it can use in place of its own training-bound beliefs about the country, including the one answer a general model will not give on its own, that the event cannot be placed. The agent asks the bridge what the model does not know. A tool-protocol server that exposes the object to such an agent exists today.

In all four patterns the integration surface is the object. Not the model, not the weights, not the pipeline. The object. A builder binds to a typed, bounded, auditable value whose contract is fixed in advance, and is thereby insulated from everything the layer does to stay current beneath it. This is the practical meaning of Section 6.3: the object is the product, and it is also the seam along which other products attach.

8. Country-agnosticism

8.1 India is the first implementation, not the point

The founder's claim again:

> "LLMs cannot be country agnostic, and we need an intelligence layer that has no training cutoff date, rather they are trained every day."

A general model cannot be the right model for every country at once, because being right for a country means holding that country's structure, and a model that averages over all of them holds none of them exactly. The answer proposed here is not a better general model and not one model per country. It is a layer whose machinery is the same everywhere and whose content is written country by country.

Everything in this paper so far is about India because India is where the content was written. Nothing in the machinery is about India.

8.2 What is the same everywhere

Part of the architecture	Why it does not depend on the country
The order of trust in event resolution	Evidence in the text outranks a model's guess in any language and any market.
Refusal as a stored outcome	Every country has events the layer has not been taught.
The review loop	Knowledge needs a human approver wherever it is written.
The outcome monitor	A claim that an event moves a sector can be checked against that country's sector returns.
The structured object	Event, signed sector vector, stakeholder views, regime, audit.
Ingestion rules	One fetcher, one store and one client per source, alerts on empty runs, labelled writers.

8.3 What every country must write for itself

Content	What it holds	Why it cannot be borrowed
Sector schema	The sectors in which incidence is expressed	Economies are divided differently, and listed proxies differ.
Event taxonomy	The events that matter, their direction and exposure	A tax, a subsidy or a monsoon is an event in one country and not in another.
Rules	Chains with delays	The chain from rain to a price runs through that country's crops, roads and policy.
Pressure to sector matrix	How each pressure lands on each sector	Import dependence and subsidy structure differ.
Context library	Festivals, fiscal dates, harvests, standing geopolitical exposures	A calendar is a country's own.

Content	What it holds	Why it cannot be borrowed
Sources	The official feeds and their formats	Regulators, exchanges and ministries are national.
Reviewers	The people who approve	Judgment about a country belongs to people who know it.

The context library already shows the pattern in miniature. It is organised by kind of phenomenon with the region as a label, and it holds entries for regions other than India where they bear on India: the factory shutdown around Chinese New Year, and the Strait of Hormuz. Its design note says that adding a region is an edit to data, not to code.

8.4 The same rain, two effects

The founder's example of why content cannot be borrowed, even inside one country:

> "The same rainfall that damages yield in Punjab by 20% would have double the effect in Northeast, because the infrastructure in Northeast is fragile."

This is an illustration of a design requirement, not a measured result. No rule or evaluation in the present system measures a regional multiplier of this kind, and the figures are the founder's. What the example establishes is the shape of the knowledge that is needed. The event is the same word in both places. The incidence is not, because incidence is the event multiplied by the capacity of the place to absorb it: storage, roads, alternative routes, irrigation, the speed of relief. A layer that records "heavy rain reduces yield" has recorded the event. A layer that records where the roads fail has recorded the incidence.

The public record supports the direction of the founder's point, though the specific multiplier is his and unmeasured. Punjab irrigates about 98 to 99 percent of its net sown area, the highest share in India, so a rainfall shortfall there meets stored water, assured irrigation and a strong procurement system. Across the Northeast more than sixty percent of cultivated land is rain-fed, only about 14 to 23 percent is under assured irrigation, and the terrain carries heavy flood and landslide exposure, so the same shock meets far less capacity to absorb it. This is why the contextual layer is organised by region for a country as varied as India: the event is one word in both places, and the incidence is set by the irrigation, storage, roads and relief that differ between them. The sources are listed at the end of the paper.

It follows that country is not the finest useful grain. The present system carries region as a field on some rules and as a label in the context library. It does not yet model regional absorption in general, and Section 10 lists that as a limit.

8.5 A minimal specification for a new country

Appendix B gives the specification in full. In summary, a first implementation for a new country needs, in this order:

1. a sector schema and a source of sector returns to check claims against;
2. a list of the few macro events that move the whole market, each with a direction per sector and a stated basis;

3. three or more official sources, read daily under the ingestion rules;
4. the refusal store, switched on from the first day;
5. one reviewer with authority to approve changes;
6. a context library with the national calendar;
7. rules for the chains that the event list cannot express, added only as refusals reveal them.

The order matters. The Indian implementation built items 2 and 7 before item 4, and spent several months discovering through failures what a refusal store would have shown in the first week.

This specification is untested. No second country has been implemented. The claim that the architecture transfers is a claim about design, supported by the fact that no component above refers to India, and not yet by a second deployment.

8.6 Why fine-tuning a general model does not do this

One might fine-tune a general model on a country's text and expect the same result. The development record is, in effect, a test of that expectation, and gives five reasons against it.

- Weights have a date. A fine-tuned model knows the country up to its last training run. The layer described here was last retrained in September 2026 and has changed its knowledge several times since without touching a weight.
- Weights learn the defects of their labels. The model here was trained on labels from a defective matcher, and fixing the matcher "never fixed the weights". A knowledge file can be corrected in one line; a weight cannot.
- Weights cannot hold a decision not to know. On 5 October a tariff event was held back for want of an honest anchor. There is no place in a set of weights to record that a value is being withheld on purpose.
- Weights cannot be audited by the person who must answer for them. A reviewer can read a rule and its stated source. No reviewer can read a gradient.
- Weights drift toward agreement with themselves. The retrained model was a distillation of the rule system. Left to train on its own outputs it would have confirmed them.

Fine-tuning is used inside the layer, for the two models that assist event resolution. It is used as a servant of the written knowledge, in last place in the order of trust.

8.7 The architecture as a base layer for any vertical

The same split that holds across countries holds across domains. The mechanism is domain-agnostic; the content is domain-specific. This is a design property, not a product roadmap.

The mechanism that does not change: an event taxonomy, a rule layer, the primitive pressures, the arbiter that ranks evidence above inference in a fixed order of trust, the refusal branch, the human review loop, the outcome monitor, and the one structured object they produce. None of these names a domain. Each is a statement about how events become decisions, not about which events or which decisions.

The content that changes with the domain: the sector schema, the events, the rules, the pressure-to-sector matrix, the context library, the sources, and the reviewers. Each of these describes one domain and cannot be borrowed from another, for the same reason a country's content cannot be borrowed from another country (Section 8.3).

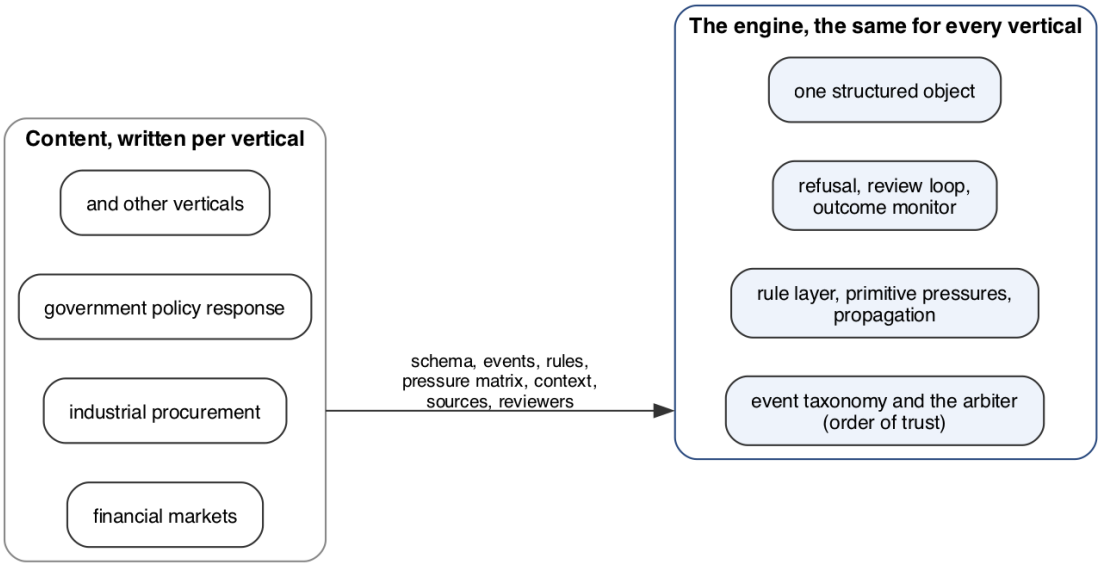


Figure 4. The same engine, different content. The mechanism is written once; each vertical supplies its own schema, events, rules, pressure matrix, context, sources and reviewers.

The domains are not only financial. The same mechanism applies to financial markets, industrial procurement, and government policy response, among others. In each, something happens, it redistributes gain and loss across a different set of actors through different channels with different delays, and a decision-maker needs the incidence before the market does.

A worked example from industrial procurement shows the same mechanism with different content. It is illustrative, in the manner of Section 8.4, and not a measured result. A purchase manager at a manufacturer reads an event: a major port on an import route announces a multi-week congestion backlog. The finance-side version of this reasoning is the apple story of Section 1. The procurement version asks the same four questions, who gains, who loses, through which channel, and when, and receives the same structured object with a different taxonomy.

Channel	Who is on the gaining side	Who is on the losing side
Longer lead times on the affected route	Holders of inventory; suppliers on alternative routes	Lines that run to a just-in-time schedule
Higher landed cost on imported inputs	Qualified domestic suppliers of the same input	Products whose margin cannot absorb the input cost
Scramble for substitutes	Second sources already qualified	Inputs with a single qualified source
Currency on the import bill	Inputs sourced locally	Inputs priced in a strengthening foreign

Channel	Who is on the gaining side	Who is on the losing side
		currency

The event resolves against a procurement event taxonomy, not a market one. The pressures are procurement pressures: lead time, landed cost, availability of substitutes, inventory cover, and currency on imported inputs. The incidence is expressed across input categories and supplier groups, not listed market sectors. The stakeholder views are the purchasing desk, the production planner and the finance function. The context library holds the shipping calendar and the supplier countries' own disruptions, the same factory shutdown around Chinese New Year that appears in the financial context library, read now for its effect on a bill of materials rather than on an index. The object tells the purchase manager that landed cost on a set of inputs will rise, that two of them have a qualified domestic substitute and three do not, that the effect reaches the line in a number of weeks, and that one input cannot be placed because the layer has not been taught it, which is the refusal branch doing in procurement exactly what it does in finance.

The design property is the point: the same engine, different content. This is why the architecture is a base layer and not a product. A product is one vertical's content running on the engine. The base layer is the engine and the discipline around it, the order of trust, the refusal, the review loop, the outcome monitor, and the object. Section 8.5 gives the minimal specification for bringing up a new country on the engine; a new vertical is the same act of authorship pointed at a different domain. Write the schema, the events, the pressures, the rules, the context and the sources, appoint the reviewers, switch on the refusal store from the first day, and let the stored refusals tell you what the domain has that the first event list did not.

9. Evaluation

The system should be evaluated by whether it preserves economic structure, not by whether it writes a fluent explanation or wins a scalar sentiment task.

9.1 A different output space

Dimension	A polarity model	This system
Output space	One class in positive, negative, neutral	A signed vector over 32 sectors, with event, pressures, stakeholders, regime and audit data
Primary question	What is the text's polarity?	Who gains, who loses, through which channel, after what delay?
Flip side	Structurally absent	Explicit
Fair evaluation	Sentiment classification	Sector sign, sector rank, causal path, stakeholder incidence, event-study validation

Collapsing the sector vector to one label destroys what the system exists to preserve. A higher score by a polarity model on a three-class task establishes performance on a different task. An

archived comparison from the early development, which collapsed the vector into sentiment labels generated by a general model and not validated by a person, is treated in this paper as it was in the earlier draft: kept for provenance and cited as evidence of nothing.

The closest public precedent is the Indian sector-ranking work represented by InFiNITE and BASIR, which classifies budget text into sectors and evaluates sector performance after the announcement. The task here is broader, since the input is arbitrary financial news and the output is signed. The direction of evaluation is the same: compare sector outputs with sector outcomes.

9.2 The framework by layer

Layer	Metric	Method
Event resolution	Accuracy, macro-F1, top-k accuracy, and refusal precision and recall	A labelled headline set that includes texts the system should refuse.
Primitive pressures	Precision and recall per primitive	Expert review.
Sector incidence	Sign accuracy, top-k sector accuracy, rank correlation	Sector index outcomes and human labels.
Propagation	Lag accuracy, downstream recall	Event studies and rule audit.
Stakeholder views	Validity of incidence	Expert review by stakeholder class.
Review loop	Proposal yield, regression rate, time in queue	The review queue and its outcomes.
Serving	Latency percentiles, error rate	Telemetry and load tests.

For market use, sign and rank matter more than magnitude. Magnitude calibration should follow only when sign, rank and path are stable.

Refusal is added to the first row in this version. A system that may decline must be scored on when it declines. Two errors are possible and they are not symmetric: placing a text that should have been refused gives a reader a false answer, and refusing a text that should have been placed gives the reader silence. Section 10 gives a current example of the second.

9.3 Sector sign accuracy and rank correlation

Sector incidence is the main output and should be scored as a signed ranking problem:

- sector sign accuracy: among sectors with a non-zero predicted value, did the realised excess return move in the predicted direction?
- top-k positive and negative accuracy: were the predicted top gaining and losing sectors among the strongest and weakest realised?
- Spearman rank correlation between predicted and realised sector ordering;
- flip-side detection: did the output keep simultaneous winners and losers?

A scalar sentiment model cannot attempt this benchmark, since it emits no vector. The fair comparison is the sector vector against realised sector outcomes, with scalar models recorded as not applicable.

9.4 Event-study probes

Three internal probes from May 2026 are kept from the earlier draft as diagnostic examples. They are not benchmark claims: the event list and basket construction were not frozen into a versioned artifact. The realised figures are market data. The system's outputs are given by direction only.

Date	Headline	The system's reading	What the market did	What it showed
12 May 2026	A bank cuts its growth estimate for India and expects two rate hikes	Monetary tightening. Consumer durables and automobiles adverse; financials and software mildly favourable.	Nifty 50 fell 1.832 percent. Consumer durables fell 2.092 percent and automobiles 2.951 percent. Financial services fell 1.182 percent and software 0.945 percent, both less than the index.	The adverse calls matched in absolute terms. The favourable calls appeared as relative resilience, which argues for scoring on excess return.
14 May 2026	Crude oil above 105 dollars a barrel	An energy price rise. Oil and gas and energy favourable; aviation, textiles, chemicals and transport adverse.	Nifty 50 rose 1.183 percent. The oil and gas basket fell 3.217 percent on the day; energy fell 0.707 percent.	A useful miss. The oil and gas bucket mixed upstream gainers with refiners under margin pressure, and the realised basket was dominated by the second group. The failure called for splitting the sector.
13 May 2026	Import duty on gold and silver doubled	A trade duty change. Manufacturing and capital goods	Capital goods rose 1.598 percent and metals and mining 2.834	The capital goods and metals channel moved as read. Power

Date	Headline	The system's reading	What the market did	What it showed
		favourable; power mildly adverse.	percent on the day. Power rose 0.606 percent.	contradicted a weak adverse call, pointing to the need for a direct precious metals basket.

Since the earlier draft, two standing sources of outcome evidence have been added.

The first is the daily outcome monitor (Section 4.12). When it was introduced on 2 September 2026 it covered 190 links between events and sectors, across 39 events, four delays and 11 sectors. The record that introduces it also limits it: its metrics rested on 44 days of index history with heavily overlapping samples and "must not be presented as validated evidence". We repeat that limit here. The monitor is a mechanism that now exists. It is not yet a result.

The second is a body of calibration evidence built on 19 September 2026: about five years of index history, market calendars, an edge-case suite and event studies anchored on calendar dates. The entry that records it is titled as evidence only, with no change to the taxonomy, weights or model. Its detailed results are internal and are not reported in this paper.

The protocol for a publishable event study is unchanged from the earlier draft: event time and class; the emitted sector vector; realised sector returns at one, three and five days; excess return over a broad benchmark; sign accuracy, top-k hit rate, rank correlation and lag profile; controls for broad market movement, regime and confounding events; and the data path, script and commit.

9.5 Other layers

Primitive pressures should be evaluated apart from market outcomes. Returns can validate realised incidence. They do not show whether the economic decomposition was right. That needs expert review: show the headline and the activated pressures, ask whether each is present, absent or overstated, and compute precision, recall and disagreement. This follows the annotation discipline of IndiaFinBench.

Stakeholder views and propagation should be audited as structured hypotheses by people who know each domain: producers, intermediaries, investors, fiscal authorities and trade actors.

One downstream use gives indirect evidence. The pre-open index predictor consumes the layer's confidence and macro signal among its features. The earlier draft reported its validation over 36 sessions from 7 April to 15 May 2026: average error under one percent on each of three indices, with direction accuracy between 47 and 61 percent and range hit rates between 58 and 81 percent. Those figures describe the predictor, which has many other inputs. They are not evidence for the layer's sector vectors and are repeated only to record that the layer is in use.

9.6 The boundary: incidence, not price prediction

Layer	Claim
Event incidence	Intended capability.
Primitive pressure activation	Intended capability, subject to labelled data.
Sector redistribution	Intended capability, strongest for sign and rank before magnitude.
Stakeholder exposure	Intended capability, evaluated by expert audit.
Temporal propagation	Research capability; needs lag validation.
Regional absorption	Not yet modelled in general.
Guaranteed market prediction	Not claimed.
Trade alpha	Not claimed.
Broker or execution recommendation	Not claimed.

The rule for claims is the one the earlier draft set and the commit history enforces:

```
No public claim without dataset, metric, baseline, failure examples,
and reproduction path.
```

No comprehensive public benchmark exists for signed 32-sector redistribution from arbitrary Indian financial headlines. Building one, across event class, pressures, sector sign, sector rank, stakeholder views and refusal, remains the first research priority.

10. Limitations and what the paper does not claim

The limits of the earlier draft remain.

- Headlines that carry several events can exceed single-event resolution.
- Sparse headlines may not carry enough context for reliable incidence.
- Primitive activations need stronger labelled data.
- Sector magnitude is less mature than sign and rank.
- Lag estimation is hard to validate.
- Informal-economy and regional spillovers are only partly observable in listed-sector data.
- Fallback templates improve coverage and must not become authority without review.
- The regime overlay is incomplete without wider liquidity inputs.
- Public reproducibility is limited by the commercial boundary around coefficients and weights.

The development arc adds the following.

- The origin is undocumented. The founding observation and the two years after it rest on the founder's account. The record begins in March 2026.
- Magnitudes are largely authored, not measured. The taxonomy's own header records its calibration status as pending. Events added through review borrow their magnitudes from the nearest existing event and are labelled as borrowed.

- The neural model is not independent evidence. It was trained on the engine's labels, and most of its event labels had fewer than five training examples at the last retraining. When the model and the rules agree, that is one opinion.
- Phrasing still decides too much. On 7 October 2026 the central bank raised its policy rate. The plainly worded announcements, of the form "hikes repo rate to" a level, were refused. Other headlines about the same decision were placed on the correct event at low confidence. Refusal protected readers from a wrong answer and gave them silence on the most important event of the day. The cause had not been established at the time of writing. A general model would have read every one of those headlines without difficulty, which is the argument of Section 3.4 from the other side.
- Accepted answers are still sometimes wrong. A reading of that day's accepted items found a corporate results headline placed on a seasonal consumption event, a foreign reserves headline about another country placed on the Indian event of that name, and a daily roundup of stocks in the news placed on a broad crash event. The order of trust reduces these errors. It does not remove them.
- The human is a bottleneck. More than twelve thousand refusals were awaiting a person on the date of this draft. A design that requires human approval is limited by human time, and the present queue is longer than one reviewer can clear. Triage of the queue is an open problem.
- Outcome evidence is thin. The monitor exists. Its history is short and its samples overlap.
- Silent failure has been the characteristic defect. Four times in this record a component failed without any signal: the feeds after the migration, the scheduled jobs, the context layer, and the log of unplaced headlines. Each was found by a person looking, not by a test. More such failures should be assumed to exist.
- Several renderings are designs. Of the readers described in Section 7, three are served by the current implementation: the developer, the dashboard, and the pre-open predictor.
- The context layer is not in the public response.
- Country-agnosticism is untested. No second country has been implemented.
- Regional absorption is not modelled in general. The Punjab and Northeast example states a requirement, not a result.
- The knowledge is current daily; the weights are not. "Trained every day" is true of the layer's knowledge files and queues, and not of its neural models.

The paper does not claim that the system predicts prices, that its sector vectors are calibrated in magnitude, that its causal chains are proven, or that the stack removes the need for an analyst. It claims that the unit of analysis is right, that the architecture around it is the minimum that survived contact with real text, and that the record of how it failed is as much a contribution as the design.

11. Conclusion

The system described here is not a classifier. A classifier assigns a label and stops. This system identifies an event, and then answers the questions that the event raises in one particular country: who gains, who loses, through which channel, after what delay, under which conditions, and whether it knows enough to say.

It began with a short apple harvest and a person who followed the consequences further than the news did. The two years of reasoning that followed, by his account, and the seven months of recorded engineering after that, produced three conclusions.

First, the unit is incidence. An event has two sides, and a representation with one side loses the part that a decision needs.

Second, the knowledge is national, local and perishable. It belongs in files that a person can read, correct and date, not in weights that carry the defects of their training and the date of their last run.

Third, no part is enough alone. The general model supplies language and reasoning and has no country. The contextual layer supplies the country and reads only what it has been taught. The human supplies judgment and has only so many hours. Together they compress what the founder called "10 hours of critical thinking into 1 answer", and they can say when the answer is that they do not know.

The layer is what makes a general model useful for a country-specific decision. It is the bridge: the span that a general capability must cross to become a decision made in a particular country and, as Section 8.7 argues, a particular industry. India is where it was first written. The claim of this paper is that every country that wants its decisions informed by its own structure will have to write one, and that the machinery for doing so need not be invented twice.

Appendices

The appendices are in `paper_appendix.md` in this folder:

- Appendix A: the development timeline (full version with evidence paths in `timeline.md`; gaps in `timeline_unknowns.md`).
- Appendix B: the minimal country specification.
- Appendix C: the reader-to-output mapping.
- Appendix D: references (BibTeX in `paper_references.bib`).

Sources for the worked examples and the regional illustration

The worked example in Section 2.5 draws on public reporting and official data, consulted on 2026-10-11. The chain itself is a structured hypothesis, as the text states; the dated events and figures below are the public record it rests on.

- Wheat export ban, May 2022, after a heatwave cut the harvest and the crop estimate was revised down: CNBC, "India bans wheat exports to try and tame prices as a scorching heatwave curtailed output", 14 May 2022, https://www.cnbc.com/2022/05/14/india-blocks-all-wheat-exports-with-immediate-effect.html ; United States Department of Agriculture, Foreign Agricultural Service, "Extreme Temperatures Scorch Indian Wheat Production", 2022.
- Non-basmati white rice export ban, July 2023, and its withdrawal in September 2024 under a minimum export price: International Food Policy Research Institute, "India lifts export restrictions on rice", 2024, https://www.ifpri.org/blog/india-lifts-export-

restrictions-on-rice/ ; Global Trade Alert, "India: Export ban on non-basmati white rice withdrawn", September 2024, https://www.globaltradealert.org/state-act/88765 .

- India a net importer of wheat in 2016 and 2017 after a weak harvest, with private millers importing several million tonnes: United States Department of Agriculture, Foreign Agricultural Service, "India Grain and Feed Annual", 2017; Business Standard, "Will India become a net importer of wheat?", 2022, https://www.business-standard.com/podcast/economy-policy/will-india-become-a-net-importer-of-wheat-122101800107_1.html .
- Government wheat stocks at a sixteen-year low in 2024, and the government's statement that stocks were sufficient: S&P Global Commodity Insights, "India's wheat stocks hit 16-year low", 19 April 2024; Business Standard, "India has sufficient wheat stocks, no plans to change import duty: Govt", 13 June 2024, https://www.business-standard.com/industry/agriculture/india-has-sufficient-wheat-stocks-no-plans-to-change-import-duty-govt-124061300795_1.html .

India's standing as the world's largest rice exporter is reported by the same agencies and the International Food Policy Research Institute reference above.

The regional illustration in Section 8.4 draws on the following, consulted on 2026-10-11. The specific figures in the quoted sentence are the founder's and are not measured; these sources support the direction of the reasoning, that the same weather shock has a different incidence by region.

- Punjab irrigates about 98 to 99 percent of its net sown area, the highest share in India: Press Information Bureau, Government of India, Agriculture Census highlights, https://www.pib.gov.in/newsite/printrelease.aspx?relid=132799 ; Government of India, Ministry of Agriculture, "Land Use Statistics at a Glance 2022-23".
- Northeast India is more than sixty percent rain-fed, with only about 14 to 23 percent of cultivable land under assured irrigation, and faces severe flood and landslide exposure: Morung Express, "Drowning Fields, Failing Policies: The Northeast's Silent Agrarian Crisis", 2025, https://morungexpress.com/drowning-fields-failing-policies-the-northeasts-silent-agrarian-crisis ; CEED India, "From Floods to Drought: The 2025 Climate Story of India", 2025, https://ceedindia.org/from-floods-to-drought-the-2025-climate-story-of-india/ ; Mongabay India, "Climate variability in a flood-prone region is worsening impacts on agriculture", April 2026, https://india.mongabay.com/2026/04/climate-variability-in-a-floodprone-region-is-worsening-impacts-on-agriculture/ .

The goods and services tax example in Section 2.6 draws on the following, consulted on 2026-10-11.

- The 2025 rate rationalisation (the 56th GST Council meeting of 3 September 2025; a two-rate structure of 5 and 18 percent with a 40 percent rate for sin and luxury goods, effective 22 September 2025; the named categories that moved down and up): Press Information Bureau, Government of India, 56th GST Council press material, https://www.pib.gov.in/PressNoteDetails.aspx?NoteId=155151&ModuleId=3&reg=48&lang=2 ; ClearTax, "56th GST Council Meeting Highlights", 2025, https://cleartax.in/s/56th-gst-council-meet-news ; Tally Solutions, "New GST Rate Structure 2025", https://tallysolutions.com/gst/new-gst-rate-structure/ .

- The 2017 rollout and its compliance disruption, including the estimate that about seventy percent of small and mid-sized firms were unprepared: Al Jazeera, "Indian businesses brace for chaos amid GST tax reform", 30 June 2017, https://www.aljazeera.com/economy/2017/6/30/indian-businesses-brace-for-chaos-amid-gst-tax-reform ; CNBC, "India GST to bring short-term implementation challenges and uncertainty", 30 June 2017, https://www.cnbc.com/2017/06/30/india-gst-to-bring-short-term-implementation-challenges-and-uncertainty.html ; Hindustan Times, "70% small, mid-sized firms not prepared for GST rollout", April 2017.

A note on comparable tools and the reference implementation

The body of this paper is a design pattern. The reference implementation is also offered as a product: the structured object is leased as metered endpoints (Section 7.5), not as a terminal seat. This note positions it against two kinds of tool a reader may know. It is kept out of the argument above on purpose. The prices below are as publicly reported; neither vendor publishes an official price, prices change, and the sources are given at the end of this note.

Bloomberg Terminal. A broad terminal of market data, news and analytics for finance professionals, delivered as a per-user seat. Its price is not published; it is reported at about 30,000 US dollars per user per year, about 31,980 for a single terminal in 2025.

AlphaSense. An artificial-intelligence search and market-intelligence product over filings, earnings and expert transcripts, broker research and news, delivered as a per-seat enterprise subscription. It does not publish list prices; reported seats run roughly 10,000 to 22,000 US dollars per year, with enterprise deals far higher.

This system. Country-specific causal incidence for one event: a signed 32-sector vector, stakeholder views, regime and an audit trail, and refusal when it does not know. It is delivered not as a seat but as metered endpoints, the object itself, over an interface, a client library and a tool-protocol server, embeddable in another product. It is priced per use, not per seat: a free tier of ten requests a month, then paid tiers from about 999 rupees a month (about 12 US dollars) for 500 requests to about 4,999 rupees a month (about 60 US dollars) for 5,000 requests, roughly 144 to 720 US dollars a year, as listed on the product's public pricing page.

The distinction is one of kind, not only of price. A terminal and a market-intelligence search answer the question "what is the information?" and leave the reading to the user. This system answers a narrower question and returns the reading itself: for one event, who gains, who loses, through which channel and after what delay, in one country, as a structured object a program can act on. It is not a terminal and does not replace one; it is a layer that a terminal, a dashboard or an agent can call. What it adds is incidence that is country-specific and auditable. What it does not provide is the breadth of data, history and asset coverage that a terminal exists to give. The honest positioning is that these are complements at very different price points, not substitutes.

Sources for this note, consulted on 2026-10-11. Bloomberg does not publish an official Terminal price; the figure is reported by Neugroup, "Bloomberg Terminals: How Much More You'll Pay Next Year", and by costbench.com, "Bloomberg Terminal Pricing", 2025 to 2026. AlphaSense does not publish list prices; its capabilities are from AlphaSense's own product description and the Sacra company profile, and the reported seat range is from costbench.com and kompense.com, 2026. The description of this system is drawn from Sections 6 and 7 of this

paper, and its price and tiers are from the product's public pricing page, aionanalytics.in/subscribe/indian-market-intelligence.

Appendices to "Contextual Intelligence: A Causal Layer for Country-Specific Decisions"

Appendix A: The development timeline

The full timeline, with the file path and line for every entry and the quoted reason where one exists, is `timeline.md` in this folder. Changes that can be dated but are not explained by any source are listed in `timeline_unknowns.md`. This appendix is the condensed version that the paper's Section 5 cites. Entry numbers match the full timeline.

Entry	Date	Change	Stated reason on record
0.1	September 2024	The Himachal observation	Founder's account only. No artifact on disk.
0.2	2025-04-01	Earliest date in the stored headline corpus	Not recorded.
0.3 to 0.5	2025-12 to 2026-03	News code appears in the trading system (file creation times)	Not recorded.
1.1	2026-03-14	First commit: a three-class sentiment engine with sector map and volatility adjustment	Journal: a market sentiment engine for Indian markets.
1.4	2026-03-25	Model withdrawn for retraining	"A data labeling inconsistency was discovered in the training dataset."
2.1, 2.2	2026-03-26	First event taxonomy (136 events, file version 2.0.0); labels derived from events	"The v2 model had ~50% labeling errors in training data"
2.3	2026-03-26	A labelling file with header version 1.0.0 in the same commit	Not recorded.
3.1	2026-04-05	Multi-task model with sentiment, event, macro and 32-sector	Commit message states the upgrade only.

Entry	Date	Change	Stated reason on record
		outputs	
3.2	2026-04-05	First rule that reverses polarity by reader (currency)	"rupee strengthens -> negative sentiment (hurts exporters)"
3.3	2026-04-05	First live ingestion: news crawler publishes enriched items	Stated as a change.
3.6	2026-04-12	Sector sensitivities computed from market data on a rolling window	"Layer 4 complete - dynamic sensitivity calibration"
3.7	2026-04-25	Accuracy claims removed from the public page	"Remove unsupported accuracy claims"
4.1	2026-04-26	First rule file (version 3.0.0, 21 rules)	Not recorded.
4.2	2026-04-26	First primitive pressures and pressure to sector matrix	Not recorded.
4.4	2026-05-03	First stakeholder mapper and propagation engine	Not recorded.
4.5	2026-05-03	Public output standardised on the sector vector	Execution wording removed.
5.1	2026-05-09	First stored refusal	Not recorded.
5.2	2026-05-09 to 12	First official-source ingesters	Not recorded.
5.4	2026-05-09 to 2026-06-07	First review loop (weekly digest, accepted-rules file, gated retrain)	Reconstructed by a later audit.
5.5	2026-05-11	Dynamic fallback, marked as telemetry	Headlines with no rule and no vector.
5.8	2026-05-15	First scheduled retraining path	"Closed automation gaps across the self-evolving model loop"
5.9	2026-06-04 to 06	Renamed; engine and	Public repository

Entry	Date	Change	Stated reason on record
		knowledge files removed from tracked repository	hardening.
6.1	2026-06-14	Review approval writes to the live taxonomy and reloads the runtime	Changelog entry of that day.
6.4	2026-06-26 to 2026-07-01	Taxonomy cache overwritten; integrity guards added	Every analysis failed until found.
6.5	2026-07-16	Feeds repaired after the relocation	"one incomplete migration, not three independent regressions"
6.6	2026-07-16	Backlog analysed as a signal	"understand why 1,883 items needed approval"
6.8	2026-07-16 and 17	Cultural and context layer (file creation times)	Design note; decision date not recorded.
6.9	2026-08-08	News sources join the shared store	Not recorded.
6.10	2026-08-09	Word-boundary matching; article body reaches the classifier	Short keywords matched inside unrelated words.
6.11	2026-08-09	Re-queue loop fixed; failing scheduled jobs disabled	Jobs were failing silently.
6.12	2026-08-09	Daily re-test of refusals with explicit terminal states	The backlog "was misjudged by a broken engine"
7.1	2026-09-02	Rule and taxonomy evidence decide before the model	"let a model guess outrank a taxonomy match"
7.2	2026-09-02	Event model retrained with an explicit refusal class	Training labels had come from the defective matcher.
7.3	2026-09-02	Daily outcome monitor	Validate each event to sector claim against realised

Entry	Date	Change	Stated reason on record
			returns.
7.4	2026-09-02	Weekly proposals into a human queue; automatic promotion off	Stated in the entry.
7.5	2026-09-02	Sign convention error and dropped matrix entries fixed	"computed a sign opposite to their own rationale"
7.6	2026-09-02	Imports become representable	"Import events had no representation at all"
7.8	2026-09-19	Silent failures in the context layer and the refusal log made visible	"It failed silently."
7.9	2026-09-19	Calibration evidence built (evidence only)	Stated in the entry title.
7.10	2026-09-21	Context output removed from the public response	Recorded; reason not quoted here.
8.2, 8.3	2026-10-05	Taxonomy 2.0.19 to 2.0.24 through review; three policy-source ingesters	"the imposed variant waits for an honest anchor"
8.4	2026-10-08	Ingestion rules adopted	"on the owner's instruction"
8.6	2026-10-07	Plain announcements of a rate hike refused	Cause not established.

Version progression. Taxonomy: 2.0.0 on 2026-03-26 (136 events); 2.0.15 dated 2026-06-14; 2.0.18 dated 2026-08-09 (157 events measured on 2026-09-19); 2.0.24 on 2026-10-05 (162 events). Versions 2.0.1 to 2.0.14, 2.0.16 and 2.0.17 are not on disk. Rules: 3.0.0 on 2026-04-26 (21 rules); 3.0.5 by 2026-05-09; still 3.0.5 on 2026-09-02 with 39 rules. Versions 3.0.1 to 3.0.4, and any versions 1 and 2, are not on disk.

Appendix B: The minimal country specification

This is a specification, not a report of a second deployment. No country other than India has been implemented.

B.1 What is supplied by the architecture (do not rewrite)

1. Event resolution with a fixed order of trust: rule evidence, then vocabulary evidence, then a model that may abstain, then refusal.
2. A refusal store that records the text, its source, the candidates considered and the time.
3. A daily re-test of stored refusals with explicit terminal states, including "recognised, moves nothing".
4. A weekly drafting job that turns repeated refusals into proposals, with any borrowed value labelled as borrowed.
5. A review route in which a person approves, edits or rejects, and approval writes to the knowledge files, updates a hashed manifest and reloads the runtime.
6. An outcome monitor that compares each event to sector claim with realised, market-adjusted sector returns.
7. The structured object and its contract.
8. Ingestion under five rules: one fetcher per source, one store per source, one client per host, an alert on an empty run, a source label on every row.
9. Integrity guards: a knowledge file is trusted only if it has the structure of a knowledge file; a failed component reports its failure.

B.2 What the country must write

#	Content	Minimum for a first version	Notes
1	Sector schema	The sectors used by the national exchange's indices	Must match a source of sector returns, or claims cannot be checked.
2	Sector return source	Daily index levels for each sector and a broad benchmark	Needed before any magnitude is trusted.
3	Macro event list	The few events that move the whole market: central bank rate decisions, the main indirect tax, the budget, the main import commodity, the currency, the dominant weather system, the main foreign investor flow	"Macro events are not many." Start here.
4	Direction per sector for each event	Sign only, with a one-line rationale per sector	Signs before magnitudes. State the sign convention once and test it; the Indian record contains a sign

#	Content	Minimum for a first version	Notes
			convention read two ways for months.
5	Official sources	Three or more: central bank, exchange, finance ministry or equivalent	Read daily.
6	National calendar	Market holidays, festival periods, fiscal year dates, harvest seasons	Feeds the context library.
7	Standing external exposures	The trade routes, supplier countries and their calendars that the economy depends on	Entries are labelled by region.
8	Reviewer	One person with authority to approve and the knowledge to refuse	The scarce resource. Plan the queue around this person's hours.
9	Rules for chains	None on the first day	Add a rule only when stored refusals show a chain that the event list cannot express.
10	Regional absorption	None on the first day	Record the region on every rule that has one, so that it can be modelled later.

B.3 Acceptance checks for a first version

- Every event in the list returns a non-empty sector vector. An event that is recognised and returns nothing must be declared as moving nothing, or it will return to the queue forever.
- A text with no financial content is refused.
- A plainly worded announcement of each macro event is placed. This check exists because the Indian implementation failed it for a rate hike in October 2026.
- A short keyword does not match inside a longer word.
- The refusal store receives a row when the layer refuses, with the text readable.
- A deliberately broken knowledge file is rejected at load and the previous good file stays in force.
- A deliberately failed component produces a visible warning.
- A run that stores nothing raises an alert.

B.4 What must not be done

- Do not borrow another country's signs or magnitudes without labelling them as borrowed.
- Do not let automatic promotion write to the knowledge files.
- Do not train a model on the layer's own labels and then cite the model's agreement as evidence.
- Do not publish an accuracy figure without the dataset, metric, baseline, failure examples and reproduction path.

Appendix C: The reader-to-output mapping

Fields are those of the structured object `Y = {e, P(e), S_regime, V, C, A}` and the public contract. "Internal" means the field exists in the engine and is not in the public response on the date of this draft.

Reader	Event `e`	Sector vector	Top gaining and losing sectors	Stakeholder views `V`	Pressures `P(e)`	Regime	Context `C`	Audit `A`	Refusal state	Outcome monitor
Developer	yes	yes, whole	yes	yes	internal	yes	internal	yes	yes	no
Dashboard user	yes	no	yes	optional	no	yes	no	no	hidden as "no reading"	no
Forecasting model	no	aggregated	no	no	no	yes	no	confidence only	counted	no
Alert subscriber	yes	no	top two or three	no	no	only if abnormal	calendar note	no	decides silence	no
Chief financial officer	yes	grouped by pressure	yes	fiscal and trade views	yes	yes	yes	no	stated plainly	no
Board	weekly list	no	breadth only	investor view	no	weekly summary	yes	no	counted	yes
Comp	yes	no	no	fiscal	policy	no	no	yes,	stated	no

Reader	Event e	Sector vector	Top gaining and losing sectors	Stakeholder views V	Pressures P(e)	Regime	Context C	Audit A	Refusal state	Outcome monitor
liance or regulatory author				view	restriction only			with source	plainly	
Purchase manager	yes	own input sectors only	no	producer and trade views	import dependency, logistics, cost push	no	yes, with dates	no	stated plainly	no

Status of each rendering on 2026-10-09:

Rendering	Status
Developer payload over the hosted interface, client package and tool-protocol server	Built.
Dashboard news items with event and sector fields	Built in part.
Features for the pre-open index predictor	Built.
Alert, two-page briefing, weekly board summary, regulatory note, purchase signal	Not built. Specified here.

Rule for every rendering: it may select, order and phrase. It may not add a sector, a sign, a stakeholder or a date that the object does not contain.

Appendix D: References

The reference list of the earlier draft is kept in full. No reference has been added, because this rewrite draws its new material from the project's own record and the founder's account, not from new literature. BibTeX entries are in `paper_references.bib`. Entries there carry only the fields that the earlier draft gave; volume, pages and identifiers were not in the earlier draft and have not been supplied from memory.

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Two typographic changes from the earlier draft's list, made because this folder's rules allow plain characters only: an author's name is written "Brandas" without its diacritics, and the infinity symbol in the InFiNITE title and a long dash in the BASIR title are removed. Restore them before any publication.

Project sources cited in the text (all private, on the development machine): the three development journals of March 2026; the public and internal changelogs; the audits and reports of 16 July and 9 August 2026; the status, integrity and architecture documents of the engine; the ingestion rules of 8 October 2026; the local copy of the project home page. Paths are in `timeline.md` and `verification.md`.